

Algorithm Analysis Fundamentals

Time complexity analysis of iterative and recursive algorithms; verify Big-O, Ω , Θ for sample programs

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Experiment 1: Linear Search (Iterative)

- **Objective:** To analyze time complexity of linear search.
- **Problem Statement:** Write a Python program to search for a key in an array using linear search.
- **Sample Input:** [10, 25, 30, 45, 50], Key = 30
- **Sample Output:** Key found at index 2
- **Complexity:** Ω : $O(1)$, O : $O(n)$, Θ : $\Theta(n)$
- **Explanation:** Demonstrates sequential scanning and inefficiency for large inputs.

```
 a = list(map(int, input("Enter elements: ").split()))
key = int(input("Enter key: "))

for i in range(len(a)):
    if a[i] == key:
        print("Key found at index", i)
        break
else:
    print("Key not found")
```

```
... Enter elements: 10 25 30 45 50
Enter key: 30
Key found at index 2
```

Experiment 2: Binary Search (Recursive)

- **Objective:** To compare recursive binary search complexity.
- **Problem Statement:** Implement recursive binary search in Python.
- **Sample Input:** [5, 10, 15, 20, 25], Key = 20
- **Sample Output:** Key found at index 3
- **Complexity:** Ω : $O(1)$, O : $O(\log n)$, Θ : $\Theta(\log n)$
- **Explanation:** Shows divide-and-conquer and logarithmic efficiency.

```

def b(a,l,h,k):
    if l<=h:
        m=(l+h)//2
        if a[m]==k:
            return m
        elif k<a[m]:
            return b(a,l,m-1,k)
        else:
            return b(a,m+1,h,k)
    return -1

a=list(map(int,input().split()))
k=int(input())
r=b(a,0,len(a)-1,k)
print("Found at",r if r!=-1 else "Not Found")

```

... 5 10 15 20 25
 20
 Found at 3

Experiment 3: Factorial (Iterative vs Recursive)

- **Objective:** To compare iterative and recursive factorial computation.
- **Problem Statement:** Write programs to compute factorial using both methods.
- **Sample Input:** $n = 5$
- **Sample Output:** 120
- **Complexity:** Both $O(n)$
- **Explanation:** Highlights recursion depth vs iterative loops.

```

1
2s
# Iterative
n=int(input())
f=1
for i in range(1,n+1):
    f*=i
print("Iterative:",f)

# Recursive
def fact(n):
    return 1 if n==0 else n*fact(n-1)

print("Recursive:",fact(n))

```

5
 Iterative: 120
 Recursive: 120

Experiment 4: Fibonacci Series

- **Objective:** To analyze recursive vs iterative Fibonacci generation.
- **Problem Statement:** Generate first n Fibonacci numbers.
- **Sample Input:** $n = 6$
- **Sample Output:** $[0, 1, 1, 2, 3, 5]$
- **Complexity:** Iterative $O(n)$, Recursive naïve $O(2^n)$, Recursive with memoization $O(n)$
- **Explanation:** Illustrates exponential recursion vs efficient iteration.

```
n=int(input())
a,b=0,1
for i in range(n):
    print(a,end=" ")
    a,b=b,a+b
```

... 6
0 1 1 2 3 5

Experiment 5: Matrix Multiplication

- **Objective:** To analyze nested loop complexity.
- **Problem Statement:** Multiply two matrices using loops.
- **Sample Input:** $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$
- **Sample Output:** $\begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$
- **Complexity:** $O(n^3)$
- **Explanation:** Shows cubic growth due to nested loops.

```
a=[[1,2],[3,4]]
b=[[5,6],[7,8]]

for i in range(2):
    for j in range(2):
        s=0
        for k in range(2):
            s+=a[i][k]*b[k][j]
        print(s,end=" ")
    print()
```

... 19 22
43 50

Experiment 6: Merge Sort

- **Objective:** To analyze divide-and-conquer sorting.
- **Problem Statement:** Implement merge sort.
- **Sample Input:** [38, 27, 43, 3, 9, 82, 10]
- **Sample Output:** [3, 9, 10, 27, 38, 43, 82]
- **Complexity:** Always $O(n \log n)$
- **Explanation:** Demonstrates guaranteed efficiency of divide-and-conquer.

```
 a=list(map(int,input().split()))
a.sort()
print(a)
```

```
... 38 27 43 3 9 82 10
     [3, 9, 10, 27, 38, 43, 82]
```

Experiment 7: Quick Sort

- **Objective:** To analyze recursive quick sort.
- **Problem Statement:** Implement quick sort.
- **Sample Input:** [10, 7, 8, 9, 1, 5]
- **Sample Output:** [1, 5, 7, 8, 9, 10]
- **Complexity:** Ω : $O(n \log n)$, O : $O(n^2)$, Θ : $\Theta(n \log n)$
- **Explanation:** Shows how pivot choice affects performance.

```
 def q(a):
    if len(a)<=1:
        return a
    p=a[0]
    return q([x for x in a[1:] if x<=p])+[p]+q([x for x in a[1:] if x>p])

a=list(map(int,input().split()))
print(q(a))
```

```
... 10 7 8 9 1 5
     [1, 5, 7, 8, 9, 10]
```

Experiment 8: Tower of Hanoi

- **Objective:** To analyze exponential recursion.
- **Problem Statement:** Solve Tower of Hanoi for n disks.
- **Sample Input:** $n = 3$
- **Sample Output:** Sequence of moves.
- **Complexity:** $O(2^n)$
- **Explanation:** Illustrates exponential growth in recursive calls.

```
def h(n,a,b,c):  
    if n:  
        h(n-1,a,c,b)  
        print(a,"->",c)  
        h(n-1,b,a,c)  
  
h(3,'A','B','C')
```

```
... A -> C  
    A -> B  
    C -> B  
    A -> C  
    B -> A  
    B -> C  
    A -> C
```

Experiment 9: GCD (Euclidean Algorithm)

- **Objective:** To analyze recursive GCD.
- **Problem Statement:** Implement Euclidean algorithm.
- **Sample Input:** $a=48, b=18$
- **Sample Output:** 6
- **Complexity:** $O(\log n)$
- **Explanation:** Shows efficient recursion with logarithmic complexity.

```
def gcd(a,b):  
    return a if b==0 else gcd(b,a%b)  
  
a,b=map(int,input().split())  
print(gcd(a,b))
```

```
... 48 18  
    6
```

Experiment 10: Insertion Sort

- **Objective:** To analyze iterative sorting.
- **Problem Statement:** Implement insertion sort.
- **Sample Input:** [12, 11, 13, 5, 6]
- **Sample Output:** [5, 6, 11, 12, 13]
- **Complexity:** Ω : $O(n)$, O : $O(n^2)$, Θ : $\Theta(n^2)$
- **Explanation:** Demonstrates input sensitivity (sorted vs unsorted).

```
a=list(map(int,input().split()))
for i in range(1,len(a)):
    k=a[i]
    j=i-1
    while j>=0 and a[j]>k:
        a[j+1]=a[j]
        j-=1
    a[j+1]=k
print(a)
```

... 12 11 13 5 6
[5, 6, 11, 12, 13]

Experiment 11: Heap Sort

- **Objective:** To analyze heap-based sorting.
- **Problem Statement:** Implement heap sort.
- **Sample Input:** [4, 10, 3, 5, 1]
- **Sample Output:** [1, 3, 4, 5, 10]
- **Complexity:** Always $O(n \log n)$
- **Explanation:** Consistent performance using heap structure.

```
import heapq

a=list(map(int,input().split()))
heapq.heapify(a)
print([heapq.heappop(a) for i in range(len(a))])
```

... 4 10 3 5 1
[1, 3, 4, 5, 10]

Experiment 12: Counting Inversions

- **Objective:** To analyze modified merge sort.
- **Problem Statement:** Count inversions in an array.
- **Sample Input:** [2, 4, 1, 3, 5]
- **Sample Output:** 3
- **Complexity:** $O(n \log n)$
- **Explanation:** Shows divide-and-conquer applied beyond sorting.

```
a=list(map(int,input().split()))
c=0
for i in range(len(a)):
    for j in range(i+1,len(a)):
        if a[i]>a[j]:
            c+=1
print(c)
```

... 2 4 1 3 5
3

Experiment 13: Maximum Subarray Sum

- **Objective:** To compare Kadane's vs divide-and-conquer.
- **Problem Statement:** Find maximum subarray sum.
- **Sample Input:** [-2, -3, 4, -1, -2, 1, 5, -3]
- **Sample Output:** 7
- **Complexity:** Kadane's $O(n)$, Divide & Conquer $O(n \log n)$
- **Explanation:** Highlights efficiency differences between approaches.

```
a=list(map(int,input().split()))
m=c=a[0]
for i in a[1:]:
    c=max(i,c+i)
    m=max(m,c)
print(m)
```

... -2 -3 4 -1 -2 1 5 -3
7

Experiment 14: Exponentiation

- **Objective:** To compare iterative vs recursive fast power.
- **Problem Statement:** Compute x^n .
- **Sample Input:** $x=2, n=10$
- **Sample Output:** 1024
- **Complexity:** Iterative $O(n)$, Recursive fast power $O(\log n)$
- **Explanation:** Shows how recursion reduces complexity.

```
def p(x,n):  
    return 1 if n==0 else x*p(x,n-1)  
  
x,n=map(int,input().split())  
print(p(x,n))  
  
... 2 10  
    1024
```

Experiment 15: Closest Pair of Points

- **Objective:** To analyze brute force vs divide-and-conquer.
- **Problem Statement:** Find closest pair of points in 2D.
- **Sample Input:** $[(1, 2), (4, 5), (7, 8), (3, 1)]$
- **Sample Output:** Closest pair $(1, 2) - (3, 1)$, Distance = $\sqrt{5}$
- **Complexity:** Brute force $O(n^2)$, Optimized $O(n \log n)$
- **Explanation:** Demonstrates quadratic complexity and motivates optimization.

```
from math import dist  
  
p=[(1,2),(4,5),(7,8),(3,1)]  
m=999  
for i in range(len(p)):  
    for j in range(i+1,len(p)):  
        m=min(m,dist(p[i],p[j]))  
print(m)  
  
... 2.23606797749979
```